

SCA-GPT: A Generation-Planning-Tool Assisted LLM Agent for Fully Automated Side-Channel Analysis on Cryptosystems

Wenquan Zhou¹, An Wang¹, Yaoling Ding*, Annyu Liu, Jingqi Zhang, Jiakun Li, and Liehuang Zhu, *Senior Member, IEEE*

Abstract—Non-invasive security constitutes an essential component of hardware security, primarily involving side-channel analysis (SCA), with various international standards explicitly mandating rigorous testing. However, current SCA assessments rely on manual expert procedures, causing critical issues: inconsistent results due to expert variability, error-prone multi-step testing, and high costs with IP leakage risks for manufacturers lacking in-house expertise. Automated SCA tools that deliver consistent, expert-level evaluations are urgently needed. In recent years, large language models (LLMs) have been widely adopted in various fields owing to their emergent capabilities. Particularly, LLM agents equipped with tool-usage capabilities have significantly expanded the potential of these models to interact with the physical world.

Motivated by these recent advances in LLM agents, we propose *SCA-GPT*, an end-to-end automated LLM agent framework tailored for SCA tasks. The framework integrates a domain-specific knowledge base with multiple SCA tools to enable retrieval-augmented generation for fully automated ISO/IEC 17825-compliant testing. As a core component of *SCA-GPT*, the expert knowledge base serves as the agent’s long-term memory, enabling precise retrieval and contextual reasoning during automated testing. We further present a domain-specific expert knowledge base construction approach and two complementary evaluation metrics.

Retrieval experiments validate the effectiveness of our knowledge base construction, achieving strong performance with 84.44% and 98.33% on two complementary retrieval quality metrics. We further evaluate the overall framework across three leading LLMs: DeepSeek V3.1, Kimi K2 and Qwen3 Coder. The evaluation uses datasets spanning six cryptographic algorithms (e.g., AES, DES, RSA, ECDSA) and deploying on four hardware platforms, including smart cards, microcontrollers, and FPGAs. Results show that DeepSeek V3.1, Kimi K2, and Qwen3 Coder achieve accuracies of 83.8%, 77.8%, and 91.4%, respectively. The framework reduces evaluation time by 95.7% on average compared with manual procedures while maintaining equivalent assessment quality and automatically generating evaluation reports. Notably, *SCA-GPT* is the first advanced LLM agent specifically designed for SCA tasks.

Index Terms—Side-channel analysis, large language model, retrieval-augmented generation, expert knowledge base, automated ISO-compliant security testing.

I. INTRODUCTION

IN recent years, large language models (LLMs) have made remarkable progress in various natural language processing tasks, thanks to their emergent capabilities. Representative

models such as ChatGPT [1], LLaMA [2], DeepSeek [3], Claude [4], and Gemini [5] have achieved widespread adoption, supported by the efforts of leading tech companies like OpenAI, Meta, DeepSeek and Google, as well as the open-source community. LLMs have been successfully applied in numerous domains, including code generation [6], vulnerability management [7], and anomaly detection [8], demonstrating strong generalization abilities. Studies have shown that LLMs possess emergent capabilities such as in-context learning [9] and chain-of-thought [10] reasoning, enabling them to solve complex tasks in zero-shot [11] or few-shot settings. Frameworks like ReAct [12] have further demonstrated their potential in human-like planning and decision-making. However, despite these impressive abilities, current LLMs still face inherent limitations, such as the inability to access up-to-date information [13], perform precise mathematical reasoning [14], [15], or flexibly leverage specialized models [16]. To overcome these challenges, the emergence of LLM agents capable of using external tools offers a transformative solution. By integrating tool use, LLMs can retrieve real-time information, execute complex operations, and interact with external systems in the physical world [17], [18]. This combination of model intelligence and tool functionality significantly expands the capabilities of LLMs, granting them near-unlimited potential to affect the real world.

Hardware and embedded-system security is critical. To ensure this security, products incorporating cryptographic modules must undergo rigorous security testing before entering the market. As the main non-invasive attack technique, side-channel analysis (SCA) extracts keys and other sensitive information by monitoring physical leakages such as power consumption, electromagnetic emissions or timing, and can break encrypted systems without defeating the algorithm [19]–[21]. Since the 1990s, researchers and industry have proposed various SCA techniques such as timing analysis, simple power analysis and electromagnetic emission analysis [22], [23]. Given this threat, international standards have made non-invasive attacks a required test item. For example, FIPS 140-3 introduced requirements for non-invasive physical attacks and explicitly states that these attacks obtain secret information from cryptographic modules via external measurements [24]; ISO/IEC 17825 further defines a black-box test vector leakage assessment (TVLA) for evaluating resistance to power and electromagnetic leakage [25], [26]. However, existing side-channel security evaluations still depend on experienced testers, customised softwares and large amounts of measurement data. The literature notes that despite various attempts at automation, security testing of embedded systems

Wenquan Zhou, An Wang, Yaoling Ding, Annyu Liu, Jingqi Zhang, Jiakun Li, and Liehuang Zhu are with the School of Cyberspace Science and Technology, Beijing Institute of Technology, Beijing 100081, China. Wenquan Zhou and An Wang are also with the State Key Laboratory of Cryptography and Digital Economy Security, Shandong University, Qingdao 266237, China.

¹Wenquan Zhou and An Wang contributed equally.

*Yaoling Ding is the corresponding author (Email: dyl19@bit.edu.cn).

remains predominantly manual [27]; in SCA such manual testing is especially time-consuming and costly, with expert resources scarce and expensive. This reliance on manual expert procedures creates critical challenges throughout the evaluation process. First, expert variability introduces inconsistent evaluation outcomes, as different experts may interpret standards differently or employ varying analysis strategies, leading to results that lack reproducibility. Second, the complex, multi-step nature of ISO/IEC 17825-compliant testing is inherently error-prone during manual execution. Third, many device manufacturers lack in-house SCA expertise and must outsource evaluations to third-party laboratories, raising concerns about intellectual property leakage while incurring substantial testing costs. These hand-crafted processes make it difficult to cover all devices and environments and impede large-scale compliance testing. Thus, although standards like ISO/IEC 17825 call for a “push-button” style leakage test [27], mature automated SCA tools that deliver consistent, expert-level evaluations remain urgently needed.

Since LLM agents are capable of tool usage, could they also be applied, as illustrated in Fig. 1, to achieve precise and fully automated ISO/IEC 17825-compliant non-invasive security evaluation?

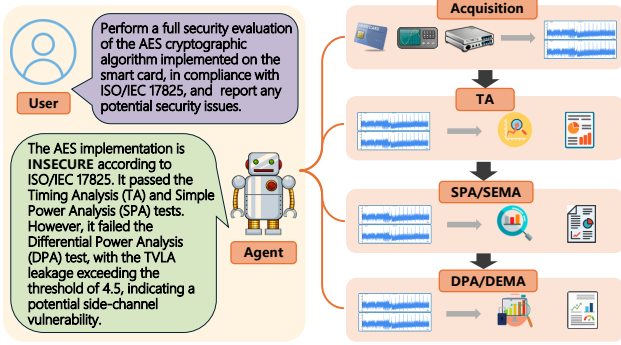


Fig. 1. LLM agents performing automated non-invasive security evaluation.

To fill this research gap, we present *SCA-GPT*, a knowledge-infused LLM agent framework designed for fully automated, non-invasive security evaluation of hardware and embedded systems. However, realizing this vision requires overcoming two fundamental challenges that prevent direct application of existing LLM technologies to the SCA domain.

First, contemporary foundation models are designed for language-to-language transformations and cannot directly process side-channel traces. While general-purpose tool-augmented agents can invoke external programs, existing agent frameworks lack specialized SCA tools required for domain-specific tasks such as power trace preprocessing, leakage quantification, and statistical distinguisher implementation. Without these purpose-built instruments, even advanced LLMs cannot bridge the gap between abstract reasoning and concrete side-channel analysis operations.

Second, effective SCA automation requires deep domain expertise that current LLMs fundamentally lack. Simply providing standards documents or research papers in the context proves inadequate due to SCA’s inherent complexity: leakage

behaviors vary dramatically across devices and implementations, and successful analysis demands context-dependent decision-making that cannot be acquired from raw documentation alone. What needed is a structured expert knowledge base that synthesizes fragmented principles from standards, literature, and expert practices into actionable reasoning patterns. The core difficulty lies in: (1) distilling coherent knowledge from heterogeneous sources spanning theoretical foundations and practical heuristics; (2) encoding context-dependent applicability rules that specify when and how to apply techniques based on device characteristics and measurement conditions; and (3) enabling precise, dynamic retrieval that surfaces relevant expertise at appropriate decision points during multi-step evaluation workflows.

To address these challenges, our key contributions are as follows:

- **Design of an autonomous, extensible LLM agent framework for SCA.** We propose *SCA-GPT*, the first advanced LLM agent tailored for SCA tasks, operating through a think-act-observe loop. This architecture enables fully autonomous execution of end-to-end security evaluations, from trace acquisition to vulnerability assessment, requiring only a single natural language instruction from non-expert users.
- **Construction of a RAG-based expert knowledge along with specialized SCA tools.** We present a domain-specific expert knowledge base construction approach that integrates standards, research literature, and handcrafted expert entries, and introduce two complementary evaluation metrics: a weighted quality score $Q@k$ that reflects the contribution of structured expert entries, and $nDCG@k$ to assess retrieval ranking quality. Together with a suite of custom-built professional SCA tools, this knowledge base provides a robust foundation for embedding domain expertise into the agent’s reasoning and enabling precise, end-to-end ISO/IEC 17825-compliant evaluations.
- **Comprehensive experimental validation across models, datasets, and platforms.** We evaluate both the expert knowledge base and the overall framework through systematic experiments. Retrieval experiments on diverse embedding models and distance functions achieve an average $Q@5$ of 84.44% and an $nDCG@5$ of 98.33%. *SCA-GPT* is further tested on three state-of-the-art LLMs: DeepSeek V3.1, Kimi K2, and Qwen3 Coder. The evaluation uses datasets covering six cryptographic algorithms deployed on four hardware platforms. The results demonstrate high accuracy (up to 91.4%), an average 95.7% reduction in evaluation time compared with manual procedures, and automated generation of evaluation reports.

II. BACKGROUND

A. Non-Invasive Security Evaluation Standard

In recent years, with the widespread deployment of cryptographic algorithms in smart cards, IoT devices, and embedded control systems, the importance of non-invasive security evaluation has become increasingly prominent. SCA enables the recovery of cryptographic keys by exploiting physical leakages without physically damaging the hardware, has emerged as a

core technique for assessing the resistance of cryptographic implementations against attacks. Timing analysis (TA), simple power/electromagnetic analysis (SPA/SEMA), and differential power/electromagnetic analysis (DPA/DEMA) are among the most commonly used SCA methods and have been successfully demonstrated against both commercial and security-grade chips.

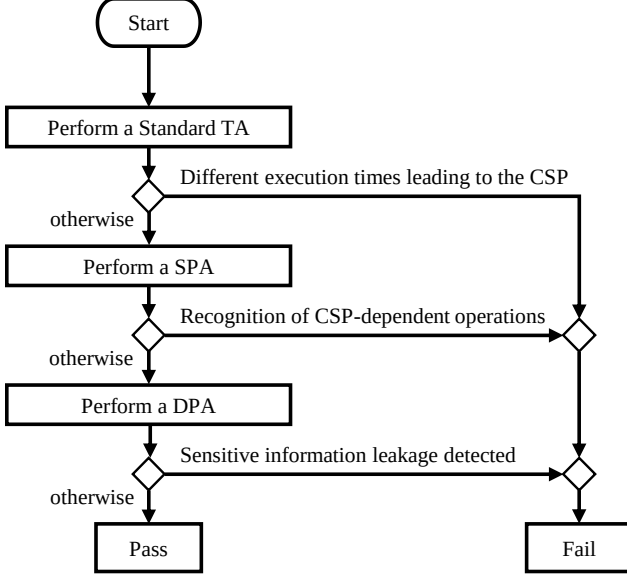


Fig. 2. Side-channel resistance test framework.

To enhance the security and compliance thresholds of cryptographic products, the International Organization for Standardization introduced ISO/IEC 17825 to unify evaluation methodologies and promote standardized testing procedures. This standard specifies a sequential testing workflow for assessing side-channel resistance (as illustrated in Fig. 2), requiring test laboratories to follow a prescribed order. For example, SPA/SEMA can only be conducted if TA has passed. Each category of attack, namely TA, SPA/SEMA, and DPA/DEMA, is accompanied by detailed testing specifications. In particular, the DPA/DEMA section incorporates the TVLA methodology, whose black-box nature significantly simplifies the detection process and reduces implementation complexity.

The adoption of ISO/IEC 17825 has not only advanced the standardization and comparability of hardware security evaluations but has also provided a unified and actionable technical foundation for secure chip design, testing laboratories, and compliance certification. However, despite these advances, conducting ISO/IEC 17825-compliant evaluations remains heavily dependent on domain experts, creating significant barriers for organizations seeking to assess their products. Existing attempts at automation either rely on semi-automatic locating of cryptographic operations in side-channel traces [28] or focus only on SPA in specific scenarios [29], still requiring substantial expert intervention throughout the evaluation process. This dependence on specialized expertise not only increases costs and time but also limits accessibility for device manufacturers who need rapid, accurate self-assessment

capabilities during development cycles.

B. LLM Powered Autonomous Agents

In recent years, LLMs have achieved remarkable progress in natural language processing and beyond, demonstrating strong generalization capabilities across tasks and domains. However, early LLMs primarily operated in static text-interaction settings, functioning essentially as “passive response systems.” Their capabilities were constrained by the information embedded in their training data and the inherent limitations of the model, making it difficult to acquire up-to-date knowledge, interact with external systems, or invoke specialized tools in real time.

To overcome these limitations, the concept of the agent was introduced. Broadly defined, an agent is an autonomous system capable of perceiving its environment, reasoning, making decisions, and taking actions based on its inputs. In an LLM-powered autonomous agent architecture, the LLM serves as the “central brain,” with its capabilities extended through several key components:

- **Tools** LLMs can call external APIs to obtain additional information that is missing from their model parameters, which are typically fixed after pretraining. The use of tools is a distinctive human trait, enabling us to transcend our physical and cognitive limitations. Equipping LLMs with external tools significantly extends their functionality and is a fundamental enabler of interaction with the physical world.
- **Memory** Memory generally include Short-term and Long-term memory.
 - **Short-term memory** Most in-context learning (prompt engineering) relies on short-term memory, which maintains temporary information within the model’s context window.
 - **Long-term memory** By leveraging external vector databases and efficient retrieval mechanisms, agents can store and recall large volumes of information over extended periods, supporting reasoning and decision-making across tasks and sessions.
- **Planning** The planning module governs the agent’s reasoning and action strategies. For instance, Chain-of-Thought [10] reasoning prompts the model to “think step-by-step,” breaking complex tasks into smaller, more manageable sub-tasks. Tree-of-Thoughts [30] expands CoT by exploring multiple reasoning paths at each step to improve decision quality. ReAct [12] integrates reasoning and action, enabling LLMs to both execute discrete task-specific actions and produce natural-language reasoning traces, forming a closed loop between thinking and acting.

Recent proof-of-concept systems, such as AutoGPT [31] and GPT-Engineer [32], have demonstrated the potential of LLM-based agents. Exploratory applications in other domains, including ChemCrow [33] for chemical research and Generative Agents [34] for simulating complex interactive behaviors, further validate the feasibility of the agent paradigm. These examples show that agent architectures hold great promise in domains requiring high decision accuracy and execution

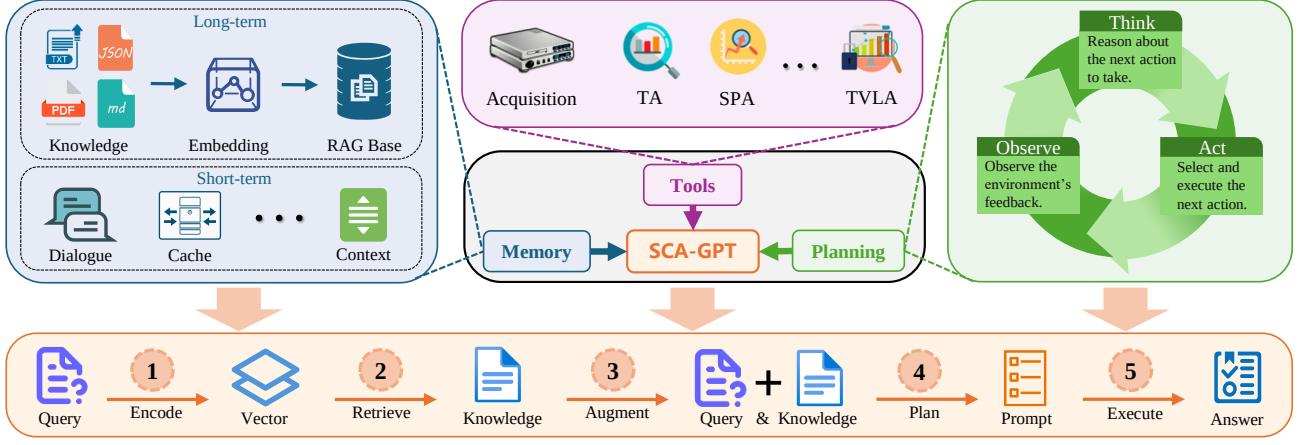


Fig. 3. The overall architecture of *SCA-GPT*. The ordinal numbers in the figure represent the order of execution.

capability, such as security testing, scientific computing, and automated operations. In the field of hardware security in particular, the autonomy and scalability of LLM-based agents lay a solid foundation for building fully automated, end-to-end security evaluation frameworks. Motivated by these advances, this work aims to leverage LLMs to address the dual challenges of tool unavailability and expert knowledge scarcity in side-channel analysis, ultimately delivering an autonomous framework capable of ISO/IEC 17825-compliant evaluations without expert intervention.

III. A GENERATION-PLANNING-TOOL ASSISTED LLM AGENT

As shown in Fig. 3, the *SCA-GPT* framework consists of three core modules working in concert to deliver automated side-channel security evaluations. The memory module (III-A) provides both short-term context tracking and long-term RAG-based expert knowledge retrieval to support continuous reasoning. The tools module (III-B) integrates a suite of specialized SCA tools for ISO/IEC 17825-compliant assessments, including trace acquisition, statistical analysis, and attack validation. The planning module (III-C) orchestrates the evaluation process through a think-act-observe loop, dynamically selecting tools and refining strategies until task completion.

The workflow proceeds in five steps, as illustrated at the bottom of Fig. 3: ① **Encode** transforms the user query into an embedding vector; ② **Retrieve** searches the RAG knowledge base within the memory module for relevant expert knowledge; ③ **Augment** integrates the retrieved knowledge with the query context maintained by the memory module; ④ **Plan** invokes the planning module to construct a complete prompt including reasoning logic, tool descriptions, and the context returned by the memory module; ⑤ **Execute** calls the selected tool from the tools module or returns the final answer. Steps ④ and ⑤ form an iterative loop where tool execution results are stored in short-term memory and fed back into the planning module, continuing until the evaluation objective is achieved.

A. Memory

The memory module comprises both short-term and long-term memory. Short-term memory holds transient information

such as the current task context, dialogue history, intermediate tool outputs, and local caches, but it is inherently limited by context-window size, volatility, and lack of curated domain structure: limitations that prevent it from reliably improving task success on its own. To address these shortcomings, *SCA-GPT* embeds a curated, domain-specific expert knowledge base into the long-term memory via a retrieval-augmented generation based mechanism. This knowledge base, constructed from standards, experiment logs, annotated traces, and literature, is model-agnostic and provides persistent, provenance-rich context that the LLM can query both at task outset and dynamically during intermediate reasoning steps, thereby materially improving planning, verification, and standards-compliant decision making. As the centerpiece of the framework, this RAG expert knowledge base is usable with any LLM; in the side-channel domain, its construction methodology and quality are critical determinants of the agent’s overall capability.

1) *RAG-based Expert Knowledge Base Construction*: The “intelligence” of an agent does not solely depend on its reasoning and planning capabilities, but more importantly on its ability to effectively leverage high-quality professional knowledge. The RAG-based expert knowledge base is therefore the core component that enables the agent to evolve from general intelligence to domain-specific expert intelligence. Moreover, the knowledge base is inherently extensible: it can be continuously enriched with new standards, experimental experience, and expert entries, progressively forming a comprehensive knowledge system for side-channel security evaluation.

We define a unified representation for all knowledge entries:

$$\mathbf{v}_i = f_\theta(E_i), \quad E_i \in \mathcal{D}, \quad (1)$$

where $\mathcal{D}$ denotes the collection of knowledge sources, including both *document-based knowledge* and *manually curated expert entries*. E_i is the i -th knowledge unit extracted from $\mathcal{D}$. The function $f_\theta(\cdot)$, typically implemented by a pre-trained embedding model such as `all-mpnet-base-v2`, maps E_i into a high-dimensional semantic space. $\mathbf{v}_i \in \mathbb{R}^{768}$ denotes the resulting semantic vector, where the dimensionality (e.g., 768 for `all-mpnet-base-v2`) is determined by the specific embedding model configuration.

Document-based knowledge includes international standards (e.g., ISO/IEC 17825), research papers, textbooks, experimental manuals, technical blogs and so on. These materials are segmented into chunks and transformed into vector embeddings by f_θ , which are then stored in the vector database.

Manually crafted expert entries are written by researchers to emphasize logic and structure. Formally, each entry is represented as:

$$E_i = \{t, [s_1, s_2, \dots, s_n]\}, \quad (2)$$

where t is the thematic label and $[s_1, s_2, \dots, s_n]$ is an ordered sequence of sub-steps. In constructing the knowledge base, the thematic label t is assigned a higher retrieval weight to emphasize its representative importance during matching. This structured formulation ensures consistency and extensibility of expert entries. For example, for ISO/IEC 17825, the entry may contain the label “ISO/IEC 17825” and a step-wise description such as “Step 1: Trace acquisition; Step 2: Timing Analysis; Step 3: Simple Power Analysis; Step 4: Differential Power Analysis.”

Ultimately, all document-based and manually curated knowledge units are represented as semantic vectors and stored in the expert knowledge base. Upon receiving a user query, the system performs retrieval over the knowledge base—commonly using Approximate Nearest Neighbor (ANN) search to identify the most relevant knowledge entries. The retrieved information is then incorporated into the prompt, which, together with the original query, is sent to the LLM for further processing and response generation.

2) *RAG-based Expert Knowledge Base Evaluation*: The utility of a RAG system hinges on two complementary abilities: (1) *capturing* the right pieces of knowledge and (2) *ordering* them so that the most useful entries appear first. We therefore adopt a dual evaluation scheme that explicitly targets these desiderata.

a) *Coverage-oriented score $Q@k$* : Given a query q , the retriever returns top- k entries $\{r_1, \dots, r_k\}$. We distinguish between expert entries and background entries. Define a gain function

$$g(r_j | q) \in \{g_E, g_B, 0\} \quad \text{with} \quad g_E > g_B > 0. \quad (3)$$

Here g_E rewards retrieving relevant expert entries, g_B rewards retrieving relevant background entries, and 0 is assigned to irrelevant entries. The coverage score is

$$Q@k(q) = \frac{1}{k g_E} \sum_{j=1}^k g(r_j | q) \times 100\%, \quad (4)$$

which lies in $[0, 100\%]$ and attains 100% when all top- k entries are relevant expert entries. Thus $Q@k$ measures whether the system *pulled the right information* into the shortlist, independently of within-list ordering.

b) *Ranking quality nDCG@k*: To evaluate ordering, we adopt normalized discounted cumulative gain with the same gain function $g(\cdot | q)$. For clarity, DCG and its normalized counterpart are presented on separate lines:

$$\text{DCG}@k(q) = \sum_{j=1}^k \frac{g(r_j | q)}{\log_2(j+1)}. \quad (5)$$

$$\text{nDCG}@k(q) = \frac{\text{DCG}@k(q)}{\text{IDCG}@k(q)} \times 100\%, \quad (6)$$

where $\text{IDCG}@k(q)$ is the maximum DCG achievable by sorting the returned entries in descending gain. $\text{nDCG}@k \in [0, 100\%]$ equals 100% precisely when the system ranks higher-gain entries (expert, then background) as early as possible. Thus $\text{nDCG}@k$ measures whether the system placed the right information at the top.

$Q@k$ and $\text{nDCG}@k$ capture distinct yet complementary aspects of retrieval: the former probes coverage of key knowledge (did expert entries surface at all?), while the latter probes ordering quality (are high-gain entries front-loaded?). The gain tiers ($g_E > g_B > 0$) make the evaluation sensitive to the expert nature of entries without committing to any particular retrieval architecture. Together, these metrics provide a concise and theoretically grounded reference point for assessing a RAG-based expert knowledge base.

B. Tools

Beyond accurate reasoning, effective evaluation also depends on precise execution. While conventional agents may generate and run code, this approach alone often leads to high error rates or repeated failures on complex programming subtasks. To mitigate this, *SCA-GPT* integrates a comprehensive suite of professional SCA tools, as summarized in Table I. These include, for example, oscilloscopes for waveform acquisition, timing analysis modules, and TVLA utilities. In addition, a Python interpreter tool is provided, enabling the LLM to implement custom analytical workflows when dedicated tools are unavailable. Guided by both the embedded expert knowledge and the toolset’s capabilities, *SCA-GPT* can interact directly with physical hardware, complete the full evaluation pipeline, from trace acquisition to leakage analysis. It produce conclusive security assessments in a fully automated manner.

TABLE I
SPECIALIZED SIDE-CHANNEL ANALYSIS TOOLS INTEGRATED INTO
SCA-GPT

Tools	Description
TVLA	Implements test vector leakage assessment for statistical detection of side-channel leakage in a black-box setting.
Retriever	Retrieves the most relevant knowledge from the expert knowledge base using ANN and summarizes the content.
Timing Analysis	Measures and analyzes execution time variations to detect timing-based leakage.
Trace Acquisition	Captures high-resolution power or electromagnetic traces from target devices for leakage analysis.
Python Interpreter	Executes Python scripts to perform custom analysis or automation tasks when no dedicated tool is available.

C. Planning

The planning module serves as the central decision-making engine of *SCA-GPT*, orchestrating task execution through a

structured *think-act-observe* loop. Upon receiving the user’s query and the expert knowledge retrieved via the RAG knowledge base, the planning module prompts the LLM to engage in iterative cycles of reasoning and action. In each cycle, the LLM first analyzes the current goal or problem context (*think*), determines the most suitable next action (*act*), and subsequently observes the feedback or output from the environment (*observe*). Actions may include invoking a specialized tool, generating and executing code, or querying additional data. This process is repeated, with each observation informing the next round of reasoning, until the task objectives are achieved and the final answer is produced.

To illustrate this workflow concretely, consider an evaluation of ECDSA implemented on a contact smart card, where the objective is to capture power traces and perform SPA. Upon receiving the user’s query, the agent retrieves relevant expert knowledge from the RAG base and sequence constraints, then enters the iterative reasoning loop:

- **Query:** Perform Simple Power Analysis on the ECDSA algorithm implemented by a smart card.
- **Prompt:** The query is augmented with retrieved expert knowledge (e.g., ECDSA double-and-add patterns, trigger configuration for scalar multiplication, clustering parameters for operation classification) and tool descriptions (acquisition tool, Python code execution environment), forming a complete prompt that guides the agent’s reasoning.
- **Execute Cycle 1:**
 - **Think:** To achieve the goal, the first step is to acquire the power traces.
 - **Act:** Invoke the acquisition tool to configure the oscilloscope, power acquisition device, and smart card reader, then execute sampling.
 - **Observe:** Acquisition completed; waveform file saved.
- **Execute Cycle 2:**
 - **Think:** With traces acquired, perform SPA by segmenting operations, applying dimensionality reduction, and clustering.
 - **Act:** Generate and execute Python code for segmentation, PCA-based reduction, and $k = 2$ clustering.
 - **Observe:** Clustering succeeded, yielding two operation classes with annotated waveforms.
- **Execute Cycle 3:**
 - **Think:** Verify sequence consistency: ECDSA’s double-and-add algorithm should not produce consecutive multiplication operations.
 - **Act:** Execute code to validate the operation sequence against expected constraints.
 - **Observe:** Consistency check passed; SPA analysis complete.
- **Execute Cycle 4:**
 - **Think:** Evaluation objective achieved; prepare final deliverables.
- **Final Answer:** Output annotated waveform plot, clustering visualization, operation sequence, and analysis summary.

This example demonstrates how *SCA-GPT* autonomously

decomposes complex evaluation tasks, leverages domain knowledge to guide decisions, and iterates through tool invocations and code generation until ISO/IEC 17825 compliance is verified.

IV. EXPERIMENT AND EVALUATION

A. Experiment Setup

The *SCA-GPT* agent framework runs on a Windows 10 Professional operating system, equipped with a 13th Gen Intel(R) Core(TM) i5-13600K CPU @ 3.50 GHz, 32 GB RAM, and an NVIDIA RTX 3080 GPU with 12 GB VRAM, providing sufficient computational capacity for large-context LLM inference, tool integration, and real-time side-channel data processing.

In our experiments, three state-of-the-art large language models are employed: DeepSeek-V3.1, Kimi-K2-Instruct-0905, and Qwen3-Coder-480B-A35B-Instruct. All models are accessed via the SiliconFlow API, with the context length set to 128K tokens to ensure consistent evaluation conditions.

The measurement setup includes a Picoscope 3403D oscilloscope for trace acquisition, an MMSCA-101 power measurement module for fine-grained signal capture, an ACR1281U smart card reader for both contact and contactless communication, and multiple test smart cards. The connection topology between the computer, oscilloscope, power acquisition module, card reader, and target device is illustrated in Fig. 4.

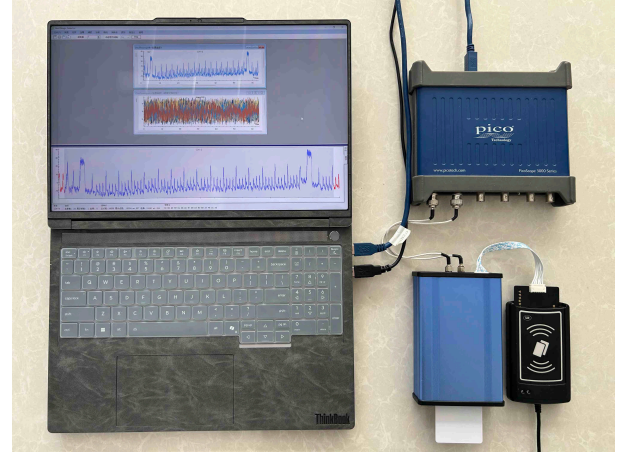


Fig. 4. Device connection.

We evaluate *SCA-GPT* on datasets spanning six cryptographic algorithms across four hardware platforms: contact and contactless smart cards (SC), STM32 microcontrollers (F103/F429), and the Sakura-G FPGA. Consistent with ISO/IEC 17825, we annotate for each dataset the leakage modality that the implementation exhibits: SPA or DPA. Under this criterion, the symmetric and hash algorithm datasets (AES, DES, SM4, HMAC-SM3) exhibit DPA leakage; the asymmetric-algorithm datasets (RSA, ECDSA) exhibit SPA leakage. Detailed configurations are summarized in Table II.

B. Construction and Validation of the Expert Knowledge Base

Building a comprehensive expert knowledge base for side-channel analysis presents several challenges: (1) the domain

TABLE II
DATASETS USED IN EXPERIMENTS

Type	Algorithm	Platform	Leakage
Symmetric	AES	Contact Smart Card	DPA
		Contactless Smart Card	DPA
		STM32F429	DPA
		STM32F103	DPA
	DES	Sakura-G (FPGA)	DPA
		Contact Smart Card	DPA
Asymmetric	SM4	Contactless Smart Card	DPA
		Contact Smart Card	DPA
	RSA	Contact Smart Card	SPA
		STM32F429	SPA
	ECDSA	Contact Smart Card	SPA
		AT89S52	SPA
Hash	HMAC-SM3	Contact Smart Card	DPA

spans multiple sub-fields (power analysis, timing analysis, fault injection, etc.) with distinct methodologies and standards; (2) critical knowledge is scattered across academic papers, international standards, and proprietary experimental logs; (3) expert entries must balance completeness with conciseness to remain effective in retrieval scenarios.

To address these challenges, we construct the knowledge base from two complementary components: document-based knowledge and manually curated expert entries.

Document-based knowledge is systematically collected and curated from four primary sources: standards (e.g., ISO/IEC 17825, GM/T series), books, academic papers spanning foundational attacks to recent advances, and technical manuals for analysis tools. These materials are segmented into chunks and transformed into vector embeddings as described in Sec. III-A1.

Manually curated expert entries are constructed following the structured format defined in Eq. (2). The construction methodology is guided by our understanding of expert reasoning in side-channel analysis: when faced with an SCA problem, human experts first identify the overall objective, then determine the execution path, and at each step along that path, they recall the specific knowledge needed for that particular sub-task. This amounts to a series of environment-dependent decisions that progressively lead to the solution. Accordingly, when constructing expert entries, we mirror this decision-making process by organizing knowledge into step-by-step sequences that are both concise and logically coherent. Each entry begins with a clear thematic label that anchors the overall goal, followed by an ordered sequence of steps that guide the reasoning process. This structure ensures that the agent can retrieve not only isolated facts but also the procedural logic that connects them, thereby enabling more effective planning and execution.

Table III summarizes the complete composition of our expert knowledge base.

Fig. 5 presents a representative expert entry for the ISO/IEC 17825 test flow.

We assess the practical utility of the expert knowledge base

TABLE III
COMPOSITION OF THE EXPERT KNOWLEDGE BASE

Category	Example	Count
Standards	ISO/IEC 17825	12
Books	Power Attack	4
Papers	Template Attacks	32
Manuals	Inspector, Detector	4
Expert Entries	ISO/IEC 17825	6
Total Knowledge Vectors		12413

```

"title": "ISO/IEC 17825 | Overall Test Flow"
"steps":
  "step": "Step 1: Perform Timing Analysis (TA)"
  "detail": "Run timing analysis to check for time-based leakages. Only if the implementation passes TA, continue to the next step."
  "step": "Step 2: Perform Simple Power Analysis (SPA)"
  "detail": "Run simple power analysis to detect visually obvious leakage patterns. Only if SPA passes, continue to DPA."
  "step": "Step 3: Perform Differential Power Analysis (DPA)"
  "detail": "Run differential power analysis to check for subtle data-dependent correlations in power traces. This is the final step."
  "step": "Step 4: Document and report results"
  "detail": "Summarize the outcomes of TA, SPA, and DPA. Clearly indicate which tests passed or failed, provide supporting metrics and explain the reasoning behind the final security assessment."

```

Fig. 5. Example of a manually curated expert entry for ISO/IEC 17825 test flow

using the dual metrics defined in Sec. III-A2: the coverage-oriented score $Q@k$ and the ranking-oriented $nDCG@k$. To make the analysis concrete and reproducible while avoiding overly system-specific confounds, we use a single canonical query, “ISO/IEC 17825”, which targets the core topic of our domain. This query stresses two desiderata simultaneously: (1) retrieving relevant expert entries and (2) ranking them above background or irrelevant entries.

We evaluate nine vector bases constructed with common embedding models and three distance functions: all-mpnet-base-v2, bge-base-en-v1.5, and gte-base-en-v1.5, each paired with cosine, dot-product, or Euclidean distance. For each base we retrieve top- $k = 5$ entries with a consistent retrieval pipeline that applies a mild, conceptually motivated LTM-aware reweighting (LTM collection emphasis and title-query alignment) and then compute $Q@5$ and $nDCG@5$ as in Sec. III-A2. The goal is not to tune a system per base, but to test whether the expert knowledge base remains effective across typical configurations.

Across configurations, $nDCG@5$ is near-ceiling (average 98.33%), indicating that once relevant entries enter the candidate set they are ranked consistently and correctly. The $Q@5$ scores are also strong (average 84.44%), showing that the system reliably surfaces critical expert entries. Notably, different distance functions and embedding families yield comparable outcomes, with a slight tendency for Euclidean variants to reach the 90% band on this query.

In practice, we view $Q@k$ as a coverage proxy (“did we surface the expert guidance?”) and $nDCG@k$ as a ranking

TABLE IV
SINGLE-QUERY EVALUATION FOR “ISO/IEC 17825” ACROSS MODELS
AND DISTANCE METRICS

Configuration	$Q@5$ (%)	$nDCG@5$ (%)
all-mpnet-base-v2 + Cosine	80.00	97.77
all-mpnet-base-v2 + Dot	80.00	97.77
all-mpnet-base-v2 + Euclidean	90.00	100.00
bge-base-en-v1.5 + Cosine	90.00	100.00
bge-base-en-v1.5 + Dot	90.00	100.00
bge-base-en-v1.5 + Euclidean	90.00	100.00
gte-base-en-v1.5 + Cosine	70.00	95.63
gte-base-en-v1.5 + Dot	80.00	93.83
gte-base-en-v1.5 + Euclidean	90.00	100.00
Average	84.44	98.33

proxy (“did we order surfaced guidance properly?”). Taken together, these results support two claims: (1) the relevant expert entries are indeed captured by retrieval, and (2) the knowledge-base structure induces stable ranking behavior across common vectorization choices. The present evidence already indicates that the expert knowledge base is effective and robust across typical retrieval configurations.

In the operational RAG corpus, thousands of heterogeneous entries including tangential or near-duplicate entries introduce substantial distractors. Under this distractor-heavy regime, the near-ceiling $nDCG@5$ and strong $Q@5$ indicate that retrieval isolates the curated expert entries from noise. This robustness follows directly from construction-time principles of the memory subsystem: (1) a unified embedding of two knowledge sources (compact, manually crafted expert entries and segmented document chunks), which creates semantically dense anchors; (2) explicit structure in expert entries that concentrates domain-salient phrases and stabilizes neighborhoods in embedding space; (3) constrained chunking of standards, experiment logs, annotated traces, and literature to preserve local co-occurrences and prevent topic bleed; and (4) provenance-preserving ingestion that reduces label noise and enforces consistent terminology across sources. Together, these design choices bias ANN retrieval toward anchor-like entries and away from semantically proximate distractors, yielding stable top- k coverage ($Q@k$) and ordering ($nDCG@k$) across embedding families and distance metrics.

C. SCA-GPT for SPA/SEMA

We take the ECDSA algorithm implemented on the contact smart card platform as an example, with the goal of capturing power traces and performing SPA. As illustrated in Fig. 6, the user only needs to issue a single-sentence query to *SCA-GPT*.

Upon receiving the user’s query, *SCA-GPT* first conducts a knowledge retrieval from the expert knowledge base, extracting key practical knowledge related to ECDSA-SPA (e.g., trigger configuration, region-of-interest selection, operation pattern identification, and sequence constraint). Operationally, once provided with the objective, expert knowledge, and tools, the agent enters a goal-conditioned *think-act-observe* loop and

Query: Perform Simple Power Analysis on the ECDSA algorithm implemented by a smart card.

Prompt: You are an expert assistant specialized in side-channel analysis. Here is relevant knowledge from your long-term memory:

Simple Power Analysis You are conducting a Simple Power Analysis (SPA) test under ISO/IEC 17825. If the algorithm is asymmetric (e.g., ECDSA or RSA), the goal is to classify these operations into two types: D and A. Follow these steps: 1. Segment the waveform into n parts. If the number is incorrect, repeat segmentation using different methods until it is accurate. 2. Apply preprocessing techniques such as moving average or filtering. 3. Perform clustering and classification to distinguish between D and A operations. Ensure that: The first operation is D. And no two A operations are adjacent.....

You can use the following tools: [JupyterAPITool, TATool, TVLATool, RAGTool.....]

Use the following format: Question: the input question you must answer. Think: you should always think about what to do. Act: the action to take, should be one of tools. Observe: the result of the action... (this Think/Act/Observe can repeat N times)

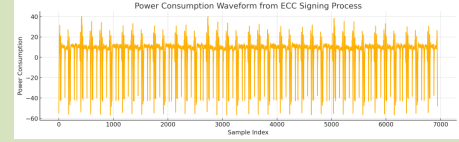
Final Answer: the final answer to the original input question.



Think 1: First, I will invoke the trace acquisition tool to automatically connect and configure the oscilloscope, power acquisition device, and smart card reader, then execute acquisition. Then, I’ll proceed to segment the trace into 36 parts. After that, I’ll classify these segments into two types: D and A. I’ll begin by invoking the trace acquisition tool and plotting the trace data.

Act 1: “TraceAcquisitionTool”

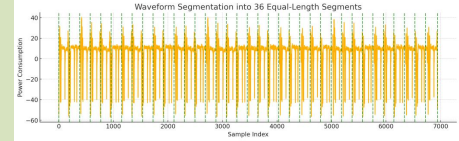
Observe 1:



Think 2: The waveform consists of 6945 data points representing power consumption during the ECDSA signing process. Next, I will segment this waveform into 36 parts. To start, I will try a simple equal-length segmentation approach, dividing the waveform into 36 segments of approximately equal length, then verify if it results in exactly 36 segments. I will plot the segmentation boundaries on the waveform for visual verification.

Act 2: “JupyterAPITool”

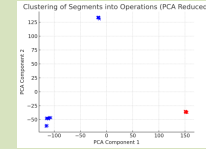
Observe 2:



Think 3: The waveform has been segmented into exactly 36 parts using equal-length segmentation. The green dashed lines indicate the boundaries between the segments. Next, I will extract each segment and classify them into two operation types: D and A. This will involve feature extraction, dimensionality reduction, and clustering, followed by visual analysis of segment shapes.

Act 3: “JupyterAPITool”

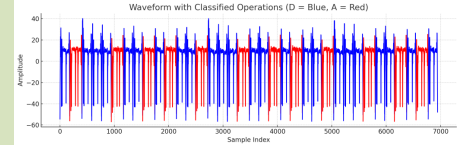
Observe 3:



Think 4: The segments have been classified into two types using PCA and KMeans clustering. Now, I will map the labels to operation types D and A, visualize them on the waveform, and output the operation sequence.

Act 4: “JupyterAPITool”

Observe 4:



Final answer: The ECDSA signing process has been successfully analyzed. The waveform was segmented into 36 operations and classified into two types. Final Sequence of Operations:

[‘D’, ‘A’, ‘D’, ‘D’, ‘D’, ‘A’, ‘D’, ‘D’, ‘A’, ‘D’, ‘A’, ‘D’, ‘D’, ‘A’, ‘D’, ‘D’, ‘A’, ‘D’, ‘A’, ‘D’, ‘D’, ‘D’, ‘A’, ‘D’, ‘D’, ‘D’, ‘A’, ‘D’, ‘D’, ‘D’, ‘A’, ‘D’, ‘D’, ‘A’, ‘D’, ‘D’, ‘A’, ‘D’, ‘D’, ‘A’, ‘D’, ‘D’, ‘A’, ‘D’]

Each operation has been visually marked on the waveform, enabling clear distinction between types D and A.

Fig. 6. *SCA-GPT* performing SPA on an ECDSA implementation.

TABLE V
ACCURACY (%) ACROSS 13 DATASETS FOR THREE LLMs

Dataset	DeepSeek V3.1				Kimi K2				Qwen3 Coder			
	Base	KB	Tools	KB&Tools	Base	KB	Tools	KB&Tools	Base	KB	Tools	KB&Tools
AES—Contact SC	9	67	74	85	29	38	79	89	0	42	78	100
AES—Contactless SC	4	38	88	94	21	37	80	81	9	35	89	98
AES—STM32F429	9	69	94	96	14	19	86	90	0	41	91	99
AES—STM32F103	7	66	78	89	0	66	51	72	0	42	55	92
AES—Sakura-G	3	52	88	97	16	47	78	92	0	52	81	100
DES—Contact SC	2	58	80	83	0	15	87	78	0	61	79	89
DES—Contactless SC	0	82	89	93	7	79	82	88	0	79	82	100
SM4—Contact SC	0	38	87	92	0	37	80	87	0	36	83	100
RSA—Contact SC	0	41	0	56	0	64	0	69	0	39	0	52
RSA—STM32F429	8	78	2	83	9	72	3	80	0	92	0	98
ECDSA—Contact SC	0	43	0	52	0	37	0	49	0	74	0	81
ECDSA—AT89S52	0	69	1	78	7	38	2	47	0	81	0	89
HMAC-SM3—Contact SC	0	33	81	91	0	33	82	90	0	48	78	90
Average	3.2	56.5	58.6	83.8	7.9	44.8	54.6	77.8	0.7	55.5	55.1	91.4

iterates until explicit stopping criteria are met. The complete experimental results are publicly available¹.

Across repeated runs on the same setup, *SCA-GPT* completed the end-to-end workflow without manual intervention. Specifically, (1) acquisition and preprocessing succeeded under default trigger templates; (2) segmentation and clustering produced two stable classes whose resulting operation sequence satisfied the ECDSA constraint; and (3) the generated report corroborated the presence of SPA leakage as defined by ISO/IEC 17825 (i.e., operations distinguishable visually or via data-driven analysis). Such professional and reliable performance was made possible primarily by the accurate expert knowledge retrieved at the beginning of the workflow.

D. SCA-GPT for Non-Invasive Security Evaluation

Following the workflow and criteria defined in Section II-A, we conduct three analyses: Timing Analysis at Level 3 with 1,000 traces using the criterion $|T_1 - T_2| < \varepsilon$; SPA to assess key-recovery capability (Section IV-C); and TVLA at Level 3 with parameters $d = 0.04$, $c = 4.5$, and 40,000 traces. All parameter choices adhere to ISO/IEC 17825 recommendations.

We evaluate 13 datasets covering six cryptographic algorithms (AES, RSA, ECDSA, SM4, DES, HMAC-SM3) across four platforms (contact and contactless smart cards, STM32 microcontrollers, and FPGA). Results are summarized in Table V. “Base” denotes the agent equipped only with a generic Python interpreter; “KB” enables the curated expert knowledge base; “Tools” enables the full domain tools; and “KB&Tools” enables both. For each dataset, we report accuracy over 100 independent runs, defined as the proportion of runs in which *SCA-GPT* correctly reports the dataset’s leakage modality per ISO/IEC 17825 (i.e., no false positives or false negatives

relative to the ground truth). The last row reports the average across all datasets. Based on these experimental results, we draw the following conclusions.

a) *SCA-GPT for symmetric algorithms*: Under ISO/IEC 17825, evaluation of symmetric ciphers primarily centers on TA and TVLA. In *SCA-GPT*, these procedures are implemented as standardized, deterministic tools, so enabling *Tools* alone typically yields higher correctness than enabling the *KB* alone. As summarized in Fig. 7, this advantage arises because *Tools* provide reproducible implementations, whereas *KB*-only depends on on-the-fly code synthesis that is susceptible to ad-hoc scripts, parameter drift, and inconsistent adherence to ISO/IEC 17825. Nevertheless, *Tools* in isolation are not sufficient: without domain knowledge, the LLM may misinterpret borderline results (such as threshold selection, leakage attribution, or dataset quality issues) and fail to draw sound conclusions. Consequently, we adopt a *KB&Tools* configuration that pairs stable computation with expert constraints and interpretive guidance. Under this recommended setting, Fig. 9 summarizes the cross-LLM average accuracy, indicating that the synergy transfers across models. Empirically, this combination delivers the most accurate and consistent outcomes on symmetric-cipher datasets.

b) *SCA-GPT for asymmetric algorithms*: Under ISO/IEC 17825, evaluation of asymmetric ciphers primarily relies on SPA. In contrast to TA/TVLA for symmetric ciphers, SPA does not admit a single standardized, deterministic toolchain; conclusions hinge on analyst choices such as segmentation and alignment, operation/opcode hypothesis testing, point-of-interest identification, and visual corroboration. Consequently, enabling *Tools* alone yields limited gains for asymmetric algorithms: generic utilities can assist with acquisition I/O, preprocessing, and visualization,

¹<https://github.com/hailife/SCA-GPT>

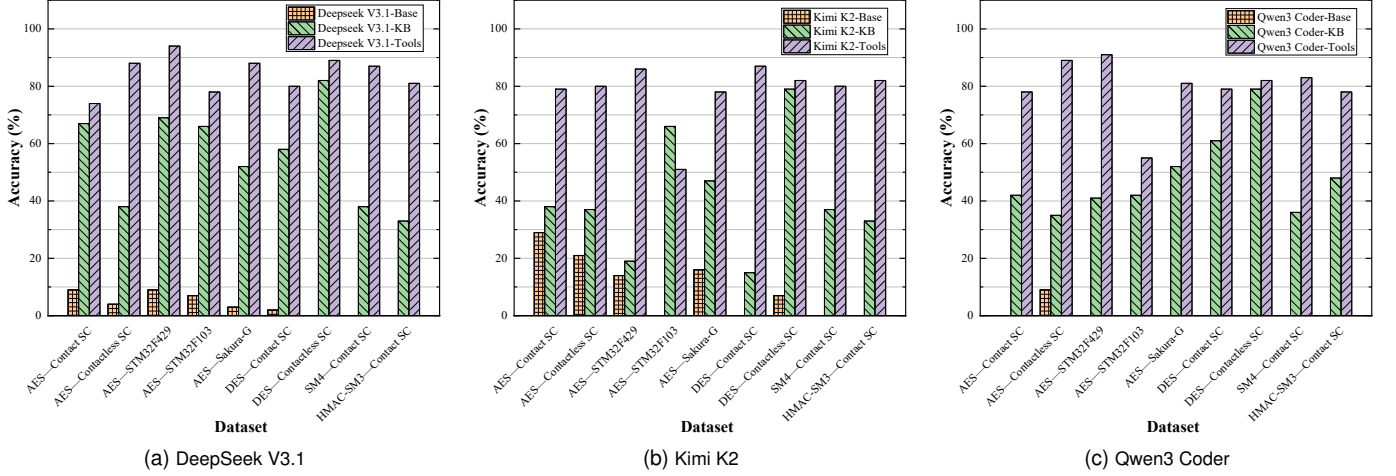


Fig. 7. Per-dataset accuracy on symmetric-cipher datasets.

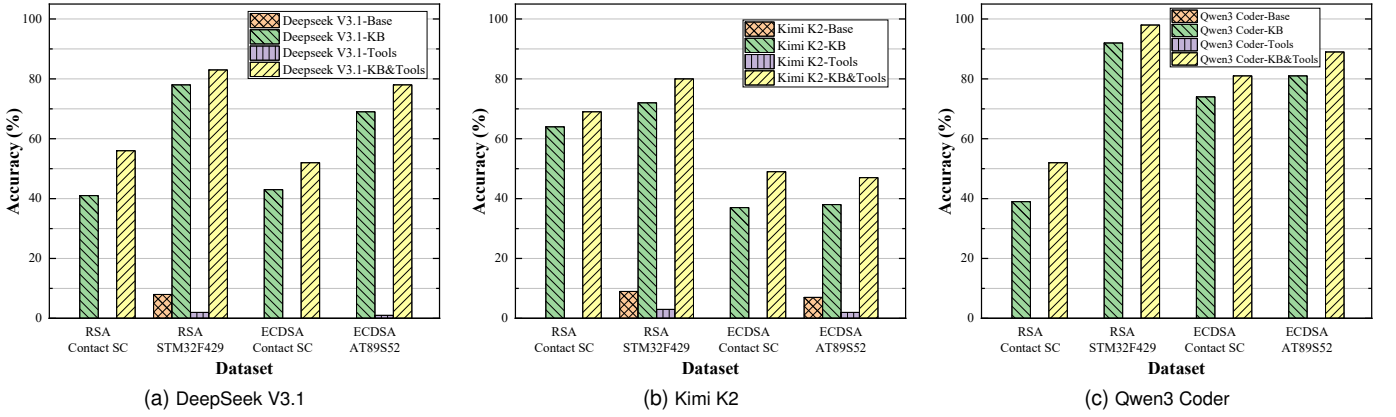


Fig. 8. Per-dataset accuracy on asymmetric-cipher datasets.

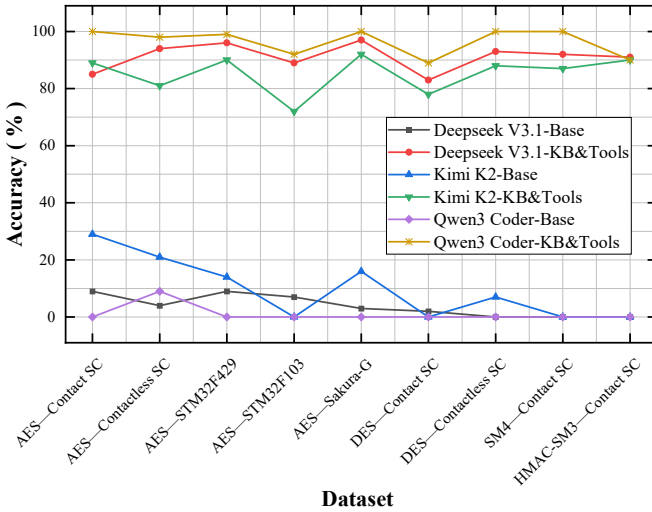


Fig. 9. SCA-GPT for symmetric algorithms.

but they cannot specify the SPA logic or adjudicate borderline cases. In *SCA-GPT*, the *KB* supplies procedure templates, expert heuristics, and acceptance criteria for SPA, from which

the LLM instantiates task-specific analysis code; thus the *KB* is the principal driver of accuracy. Combining *KB&Tools* remains beneficial: tools provide reproducible scaffolding and stable computation while the KB governs interpretation. However, the marginal improvement chiefly stems from the knowledge component. As shown in Fig. 8, per-dataset accuracy across LLMs in the asymmetric setting reflects this dependency and varies with models.

c) Cross-model generality: All three LLMs demonstrate consistent improvement trends across configurations. As summarized in Fig. 9 and Fig. 10, Qwen3 Coder performs best overall with 91.4%, followed by DeepSeek V3.1 83.8% and Kimi K2 77.8%.

Although the Qwen3-Coder has the smallest parameter count among the three LLMs considered, it attains the strongest overall accuracy when paired with KB and Tools. Once domain knowledge is available, success hinges on reliable program synthesis, tool invocation, and iterative error handling rather than raw parameter scale. The coder-oriented model translates expert guidance into correct, testable scripts, instruments the pipeline (logging/plots), and rapidly refines parameters, which collectively drive superior end-to-end outcomes.

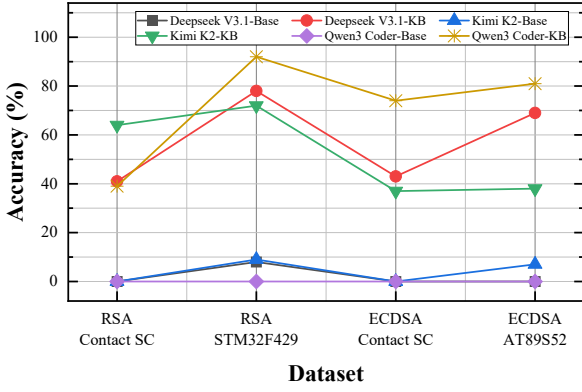


Fig. 10. *SCA-GPT* for asymmetric algorithms.

d) *Efficiency vs. manual procedures*: Finally, we compare the execution time of *SCA-GPT* with traditional manual evaluations conducted by domain experts using professional SCA software. As shown in Table VI, the framework reduces average evaluation time by 95.7%, completing typical leakage detection tasks in under 6 minutes. This significant time reduction is critical for real-world adoption in high-throughput testing environments.

TABLE VI
EVALUATION TIME COMPARISON BETWEEN *SCA-GPT* AND MANUAL

Dataset	Manual (s)	<i>SCA-GPT</i> (s)
AES—Contact SC	2685	77.75
AES—Contactless SC	3720	74.58
AES—STM32F429	2940	76.00
AES—STM32F103	2895	150.18
AES—Sakura-G	3420	88.35
DES—Contact SC	3660	130.27
DES—Contactless SC	3780	67.03
SM4—Contact SC	3435	103.23
RSA—Contact SC	4020	314.24
RSA—STM32F429	3855	230.15
ECDSA—Contact SC	3690	245.60
ECDSA—AT89S52	4230	310.19
HMAC-SM3—Contact SC	3060	74.60
Average	3491.54	149.40

V. CONCLUSION

In this work, we presented *SCA-GPT*, the first advanced LLM-based agent framework specifically designed for SCA tasks. By integrating a RAG-based expert knowledge base with a suite of professional SCA tools, *SCA-GPT* achieves fully automated, end-to-end ISO/IEC 17825-compliant evaluations. Our experimental results, conducted across three leading LLMs and diverse datasets covering six cryptographic algorithms on four hardware platforms, demonstrate high accuracy—up to 91.4%—and an average 95.7% reduction in evaluation time compared to manual procedures. Furthermore, the framework’s modular architecture ensures that it can be readily adapted to different LLMs, expert knowledge bases, and toolchains, providing strong scalability and extensibility.

Future work will focus on further enhancing *SCA-GPT*’s analytical capabilities. First, we aim to extend its evaluation

depth to support advanced CPA techniques capable of directly recovering cryptographic keys. Second, we will investigate the security of the RAG pipeline itself, particularly the risks of data poisoning and adversarially injected knowledge entries, and develop robust detection and mitigation strategies to ensure the trustworthiness of retrieval-augmented evaluations. Additionally, the open and modular nature of *SCA-GPT* allows any practitioner to refine the expert knowledge base, integrate improved tools, or adapt the framework for new evaluation standards, ensuring long-term applicability and continual evolution in response to emerging security challenges.

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